

MIAE 221: Materials Science for Engineers

Chapter 4: Imperfections in Solids (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

Instructor: Dr. Mamoun Medraj, P.Eng | **Curriculum Alignment:** Concordia University

(Aligned with Dr. Medraj Lecture 7 · Week 4)

1. 🌀 30-Second Core Intuition & Plain-English Concept

Real engineering materials are never perfect crystals. Without defects, pure metals would have theoretical shear strengths $1,000\times$ higher than observed, but they would be brittle like glass. **Defects govern all mechanical properties and atomic diffusion.**

Defects are categorized by dimensionality:

- **0D (Point Defects):** Vacancies (missing atoms), interstitials (crowded atoms), impurities.
- **1D (Linear Defects):** Dislocations (edge, screw, mixed)—the engines of plastic deformation.
- **2D (Planar Defects):** Grain boundaries, twin boundaries, external surfaces.
- **3D (Bulk Defects):** Pores, cracks, foreign inclusions.

2. ⚙️ High-Yield Mathematical Engine & Governing Laws

1. Equilibrium Vacancy Concentration (Arrhenius)

$$N_v = N \exp\left(-\frac{Q_v}{k_B T}\right)$$

- N_v : Number of vacancies per unit volume ($\text{ext}m^{-3}$).
- N : Total atomic lattice sites per unit volume: $N = \frac{\rho N_A}{A}$.
- Q_v : Energy required to form a single vacancy ($\text{ext}J/\text{atom}$ or $\text{ext}eV/\text{atom}$).
- k_B : Boltzmann's constant ($8.62 \times 10^{-5} \text{ eV/K} = 1.38 \times 10^{-23} \text{ J/K}$).
- T : Absolute temperature in **Kelvin** ($T_K = T_C + 273.15$).

2. The Four Hume-Rothery Rules for Complete Solid Solubility

To achieve complete substitutional solubility (like Cu in Ni):

1. **Atomic Size Factor:** Difference in atomic radii must be $\Delta r = \left| \frac{r_{\text{solute}} - r_{\text{solvent}}}{r_{\text{solvent}}} \right| \leq 15$.
2. **Crystal Structure:** Must have the identical crystal structure (e.g., both FCC).

3. **Electronegativity:** Electronegativities must be very similar (large ΔX forms intermetallic compounds).
4. **Valency:** A metal dissolves more of a metal of higher valency than of lower valency.

3. Weight Percent to Atom Percent Conversion

$$C_1' = \frac{C_1/A_1}{\frac{C_1}{A_1} + \frac{C_2}{A_2}} \times 100$$

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

point defect

erties of materials are profoundly sensitive to deviations from crystalline perfection; the influence is not always adverse, and often specific characteristics are deliberately fashioned by the introduction of controlled amounts or numbers of particular defects, as detailed in succeeding chapters.

A *crystalline defect* refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. Classification of crystalline imperfections is frequently made according to the geometry or dimensionality of the defect. Several different imperfections are discussed in this chapter, including **point defects** (those associated with one or two atomic positions); linear (or one-dimensional) defects; and interfacial defects, or boundaries, which are two-dimensional. Impurities in solids are also discussed, because impurity atoms may exist as point defects. Finally, techniques for the microscopic examination of defects and the structure of materials are briefly described.

Point Defects

4.2 VACANCIES AND SELF-INTERSTITIALS

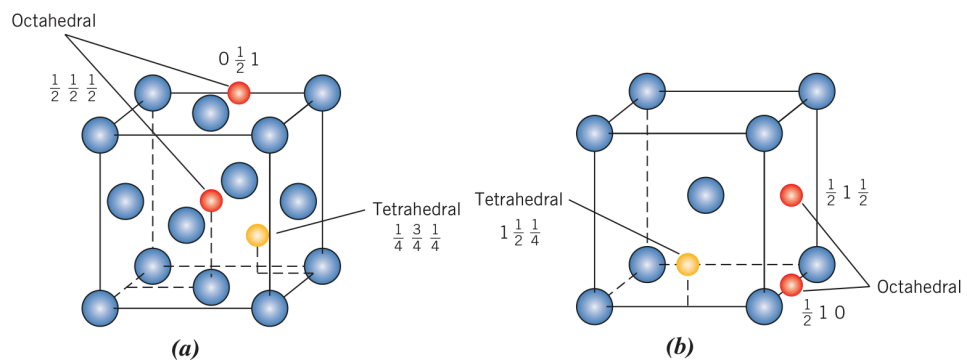
vacancy

The simplest of the point defects is a **vacancy**, or vacant lattice site, one normally occupied but from which an atom is missing (Figure 4.1). All crystalline solids contain vacancies, and, in fact, it is not possible to create such a material that is free of these

Figure 4.1: Schematic representation of a vacancy and a self-interstitial in a 2D crystalline lattice.

Figure 4.3

Locations of tetrahedral and octahedral interstitial sites within (a) FCC and (b) BCC unit cells.



have a coordination number of 4; straight lines drawn from the centers of the surrounding host atoms form a four-sided tetrahedron. However, for octahedral sites the coordination

Figure 4.3: Location of octahedral and tetrahedral interstitial voids within (a) FCC and (b) BCC crystal lattices.

In-Depth Visual Breakdown:

- **Point Defects (Figure 4.1):**
 - A **vacancy** causes surrounding lattice planes to relax inward, creating a localized tensile strain field.

- A **self-interstitial** forces host atoms severely apart, creating a large localized compressive strain field. Because self-interstitials require massive strain energy, their equilibrium concentration is orders of magnitude lower than vacancy concentration.
- **Interstitial Sites (Figure 4.3):** Shows why carbon has vastly higher solubility in FCC austenite than in BCC ferrite. In FCC, the octahedral interstitial site at the cube edge ($[1/2, 0, 0]$) has a radius ratio of $0.414R$. In BCC, the octahedral sites on faces are distorted and much smaller ($0.154R$), severely restricting carbon solubility (0.022 wt max in BCC vs 2.14 wt in FCC).

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Dr. Medraj Lecture 7 Focus:**

- **Arrhenius Linearization:** Taking the natural log of the vacancy equation:

$$\ln \left(\frac{N_v}{N} \right) = \ln A - \frac{Q_v}{k_B} \left(\frac{1}{T} \right)$$

Plotting $\ln(N_v/N)$ vs. $1/T$ yields a straight line with slope $= -\frac{Q_v}{k_B}$.

- **Burgers Vector Relationships:**

- **Edge dislocation:** $\vec{b} \perp \vec{t}$ (Burgers vector is perpendicular to dislocation line).
- **Screw dislocation:** $\vec{b} \parallel \vec{t}$ (Burgers vector is parallel to dislocation line).

- **Concordia Exam Traps:**

- **Temperature Units:** Forgetting to convert Celsius to Kelvin in Arrhenius calculations ($T = 800^\circ\text{C} \Rightarrow 1073.15\text{ K}$).
- **Hume-Rothery Partial Solubility Trap:** If an alloy satisfies 3 out of 4 Hume-Rothery rules (e.g., Cu-Zn: $\Delta r < 15$, similar electronegativity, but Zn is HCP while Cu is FCC), it exhibits **partial** (limited) solubility, NOT zero solubility!

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

Calculate the equilibrium number of vacancies per cubic meter in copper at 1000°C . Given: $Q_v = 0.90\text{ eV/atom}$, $\rho_{\text{Cu}} = 8.40\text{ g/cm}^3$ (at 1000°C), and $A_{\text{Cu}} = 63.55\text{ g/mol}$.

- **Step 1: Convert Temperature to Kelvin**

$$T = 1000 + 273.15 = 1273.15\text{ K}$$

- **Step 2: Calculate Number of Atomic Sites N per m^3**

$$N = \frac{\rho \cdot N_A}{A} = \frac{(8.40 \times 10^6\text{ g/m}^3)(6.022 \times 10^{23}\text{ atoms/mol})}{63.55\text{ g/mol}} = 7.96 \times 10^{28}\text{ atoms/m}^3$$

- **Step 3: Calculate the Arrhenius Factor**

$$\frac{Q_v}{k_B T} = \frac{0.90\text{ eV}}{(8.62 \times 10^{-5}\text{ eV/K})(1273.15\text{ K})} = \frac{0.90}{0.1097} = 8.204$$

$$\exp(-8.204) = 2.735 \times 10^{-4}$$

- **Step 4: Calculate Vacancy Concentration N_v**

$$N_v = N \exp\left(-\frac{Q_v}{k_B T}\right) = (7.96 \times 10^{28})(2.735 \times 10^{-4}) = 2.18 \times 10^{25} \text{ vacancies/m}^3$$
